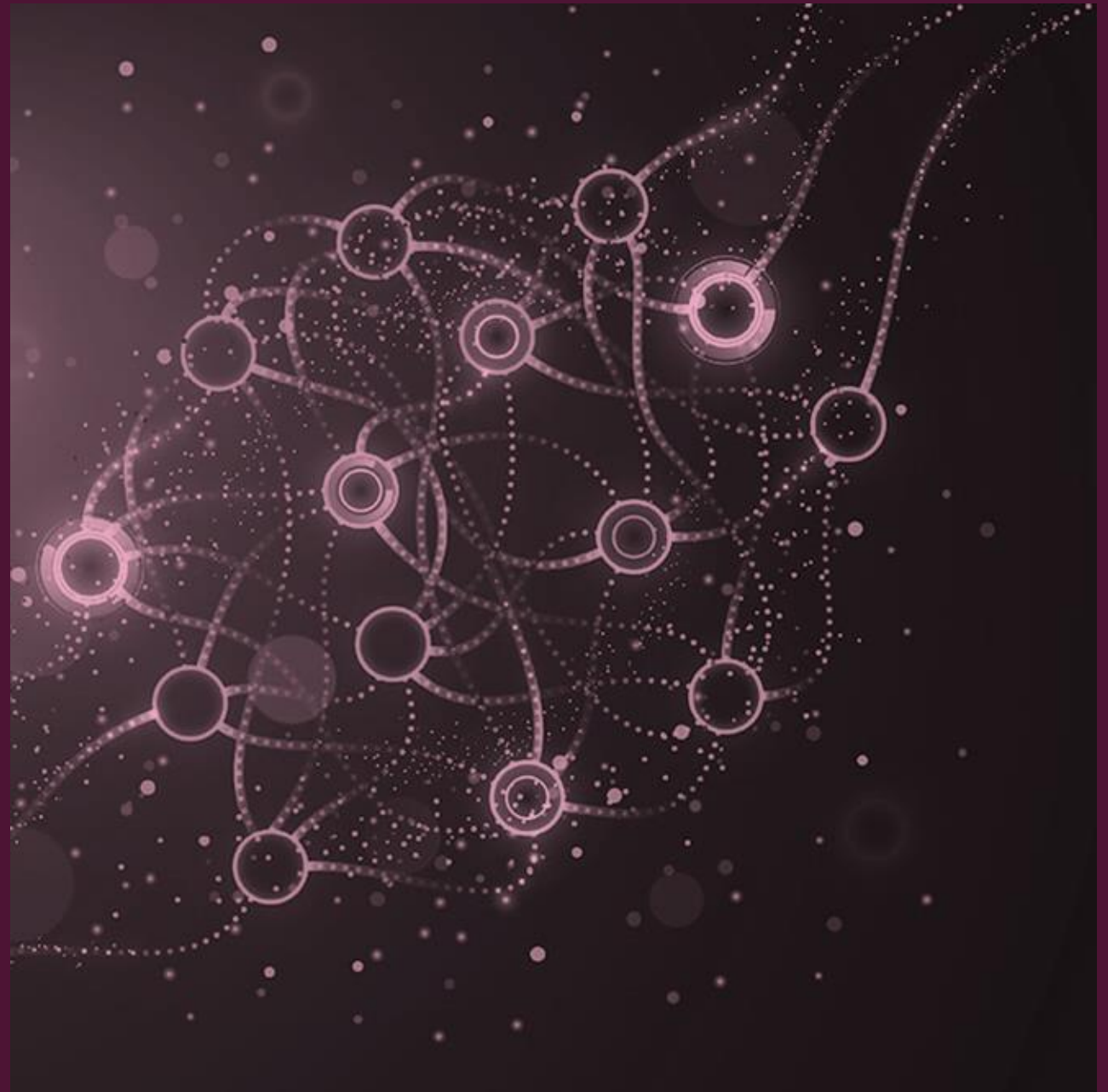
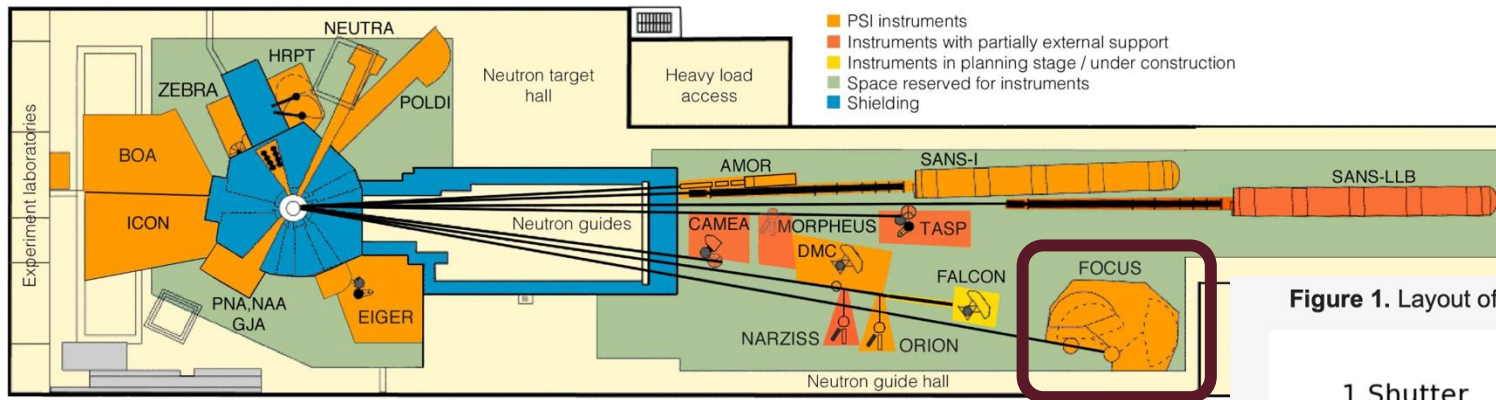


GROUP 1 PROJECT:

ISOLATING THE
INCOHERENT AND
COHERENT
SCATTERING
CONTRIBUTIONS



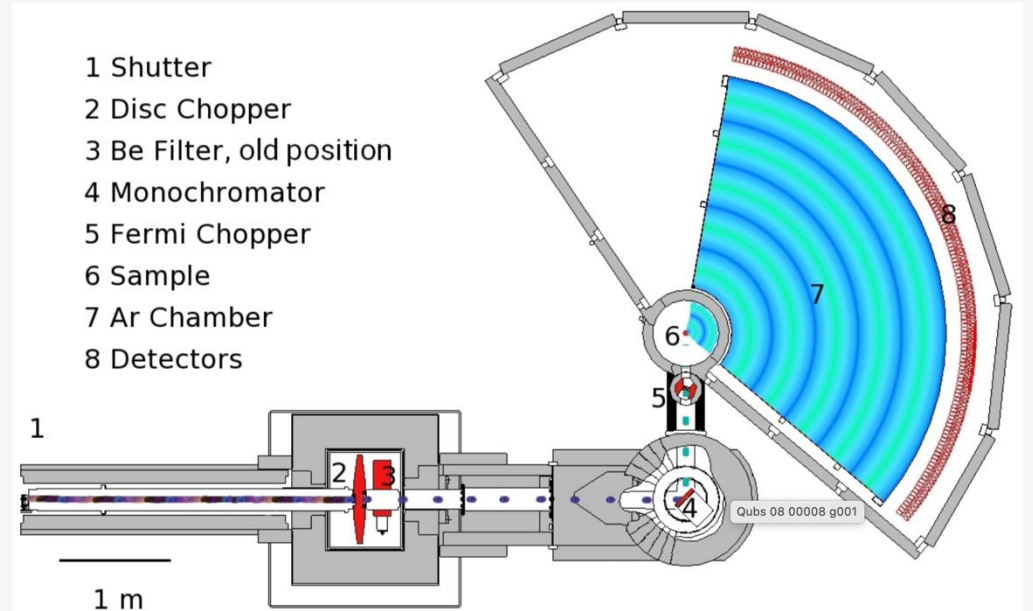
INSTRUMENT: FOCUS



SINQ floorplan (2020+) after the guide upgrade program



Figure 1. Layout of the time-of-flight spectrometer FOCUS.



SAMPLES AND PARAMETERS

5 Samples

- β -Hydroquinone;
- Liquid water;
- Copper;
- Cuprite;
- Sodalite.

SAMPLES AND PARAMETERS

5 Samples

- β -Hydroquinone;
- Liquid water;
- Copper;
- Cuprite;
- Sodalite.

Parameters

Temperature
range = 100 to 300 K
step size of = 10 K

McStas



on Juliet (CPUs)

SAMPLES AND PARAMETERS

5 Samples

- β -Hydroquinone;
- Liquid water;
- Copper;
- Cuprite;
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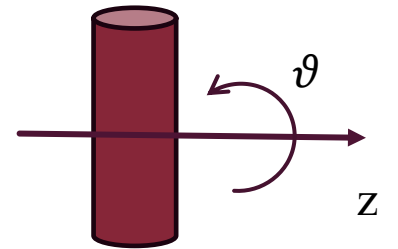
Parameters

Angle ϑ :
range = 0 to -30°
step size of = 0.33°

McStas

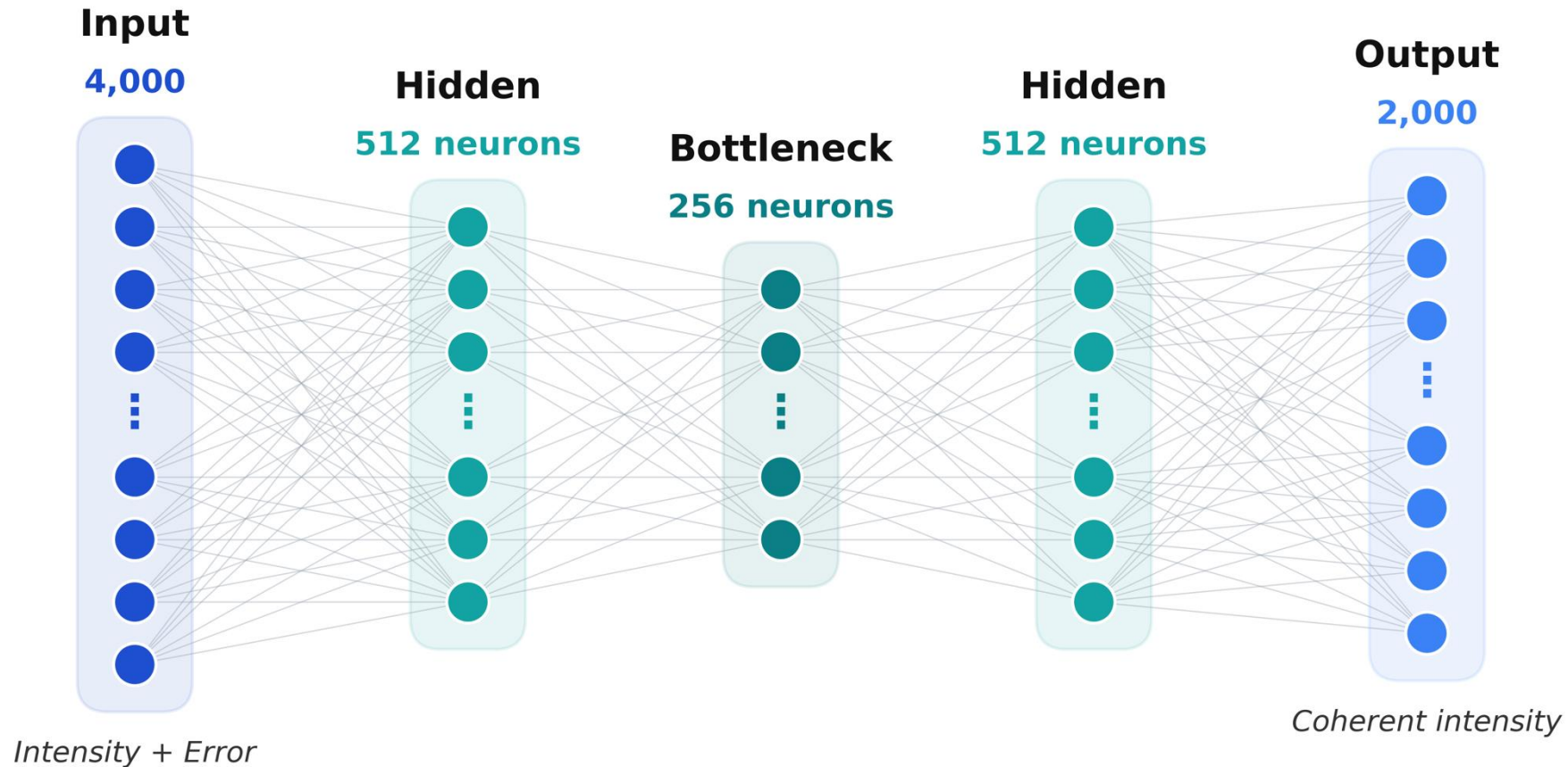


on Juliet (CPUs)

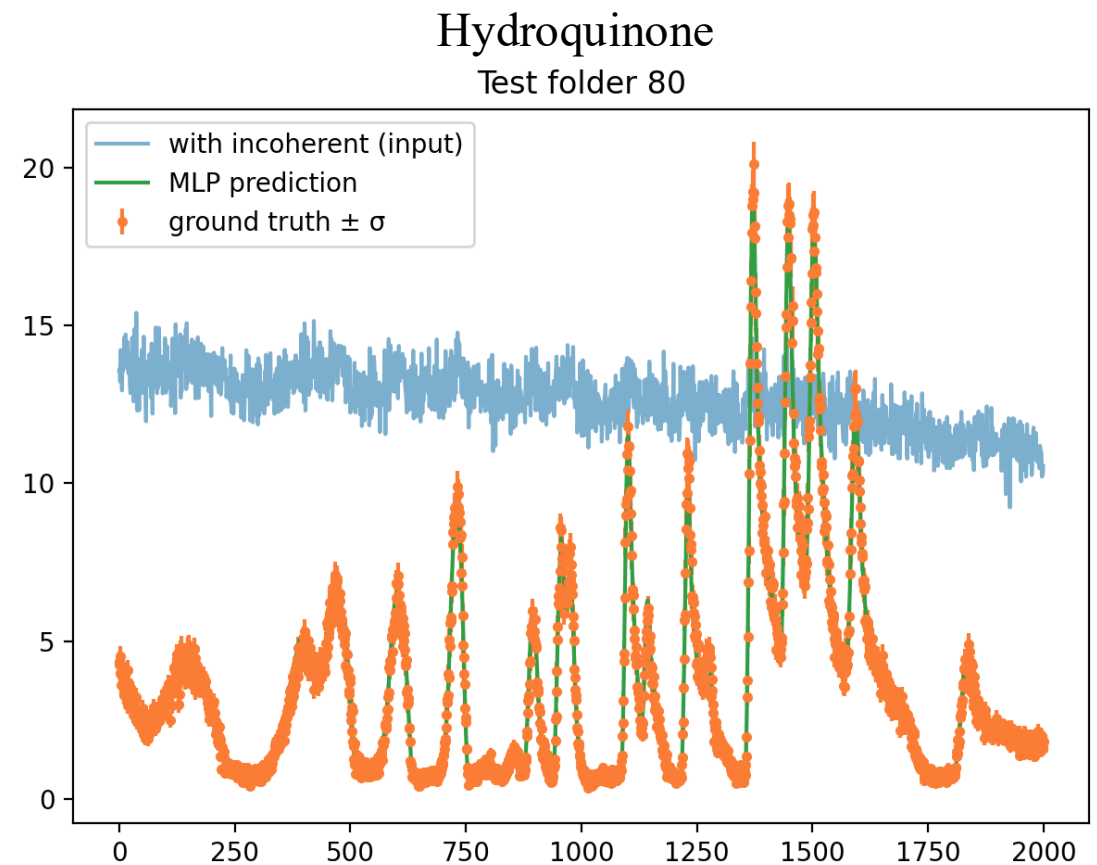
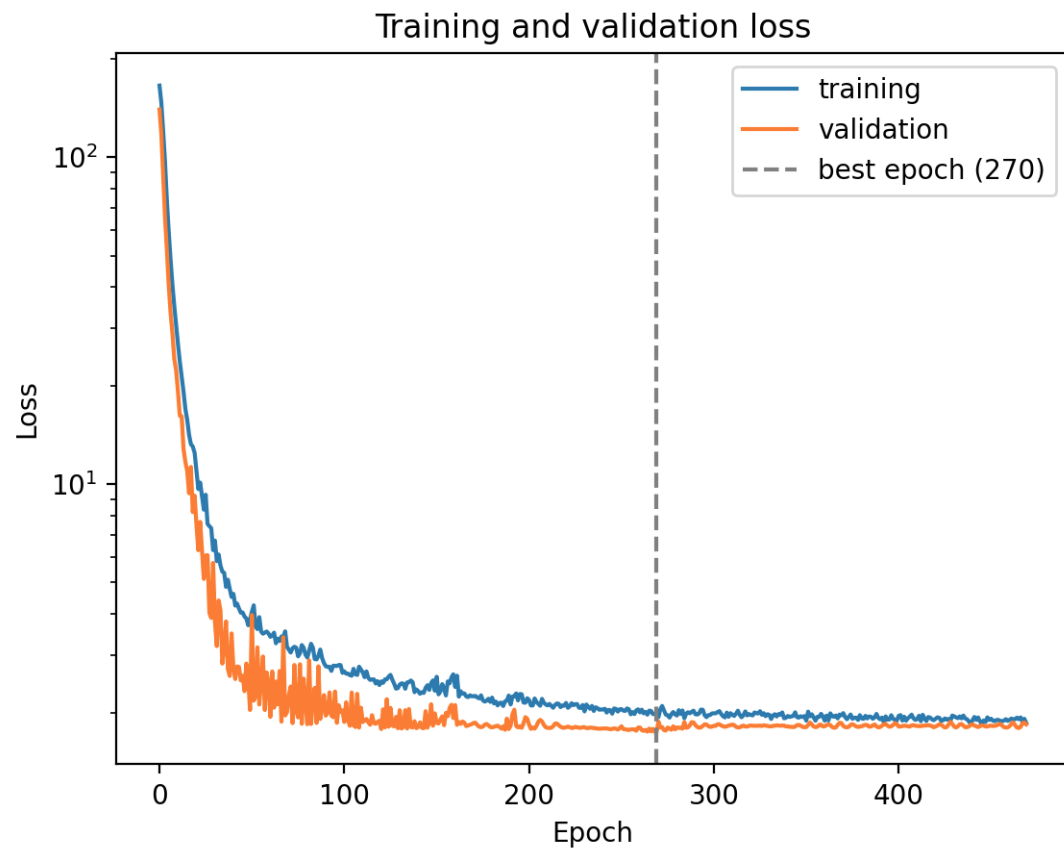


UNDER THE HOOD: THE MULTILAYER PERCEPTRON

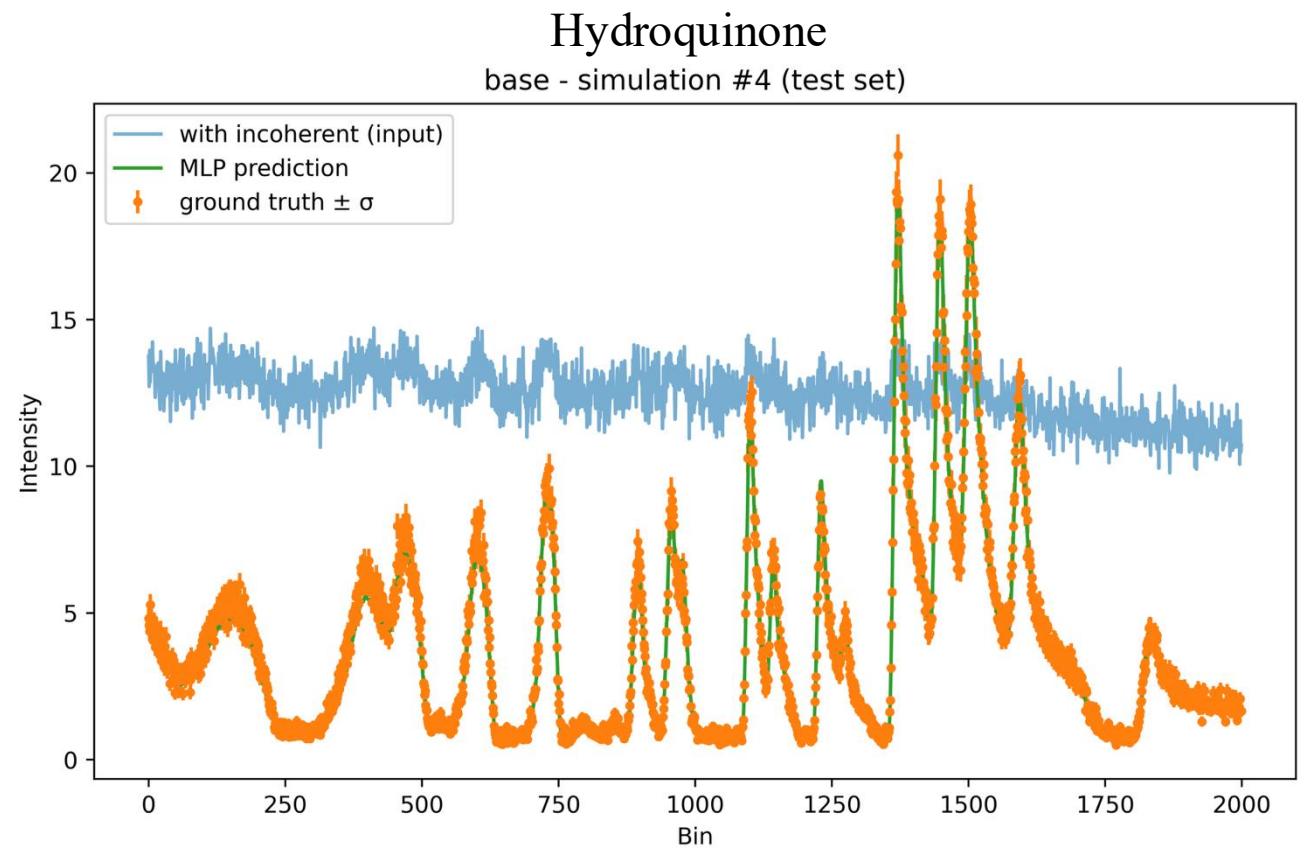
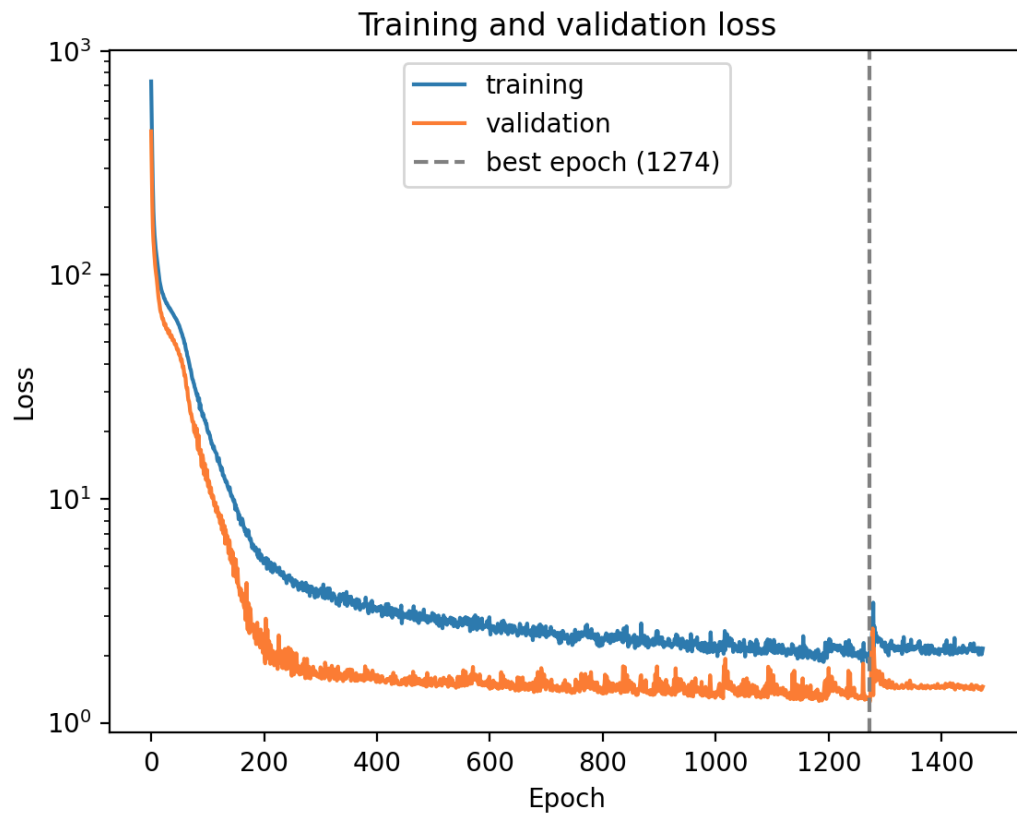
MLP for Incoherent Scattering Correction



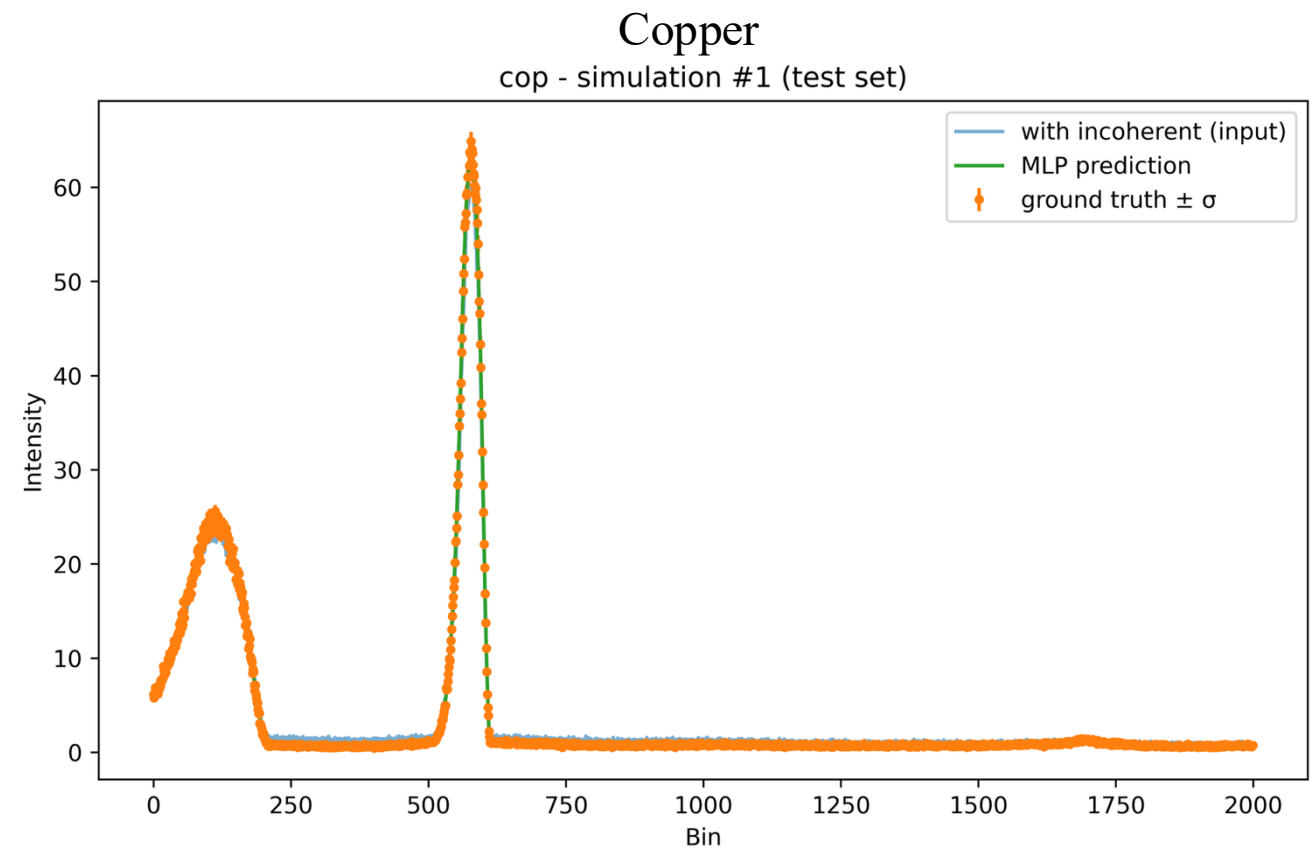
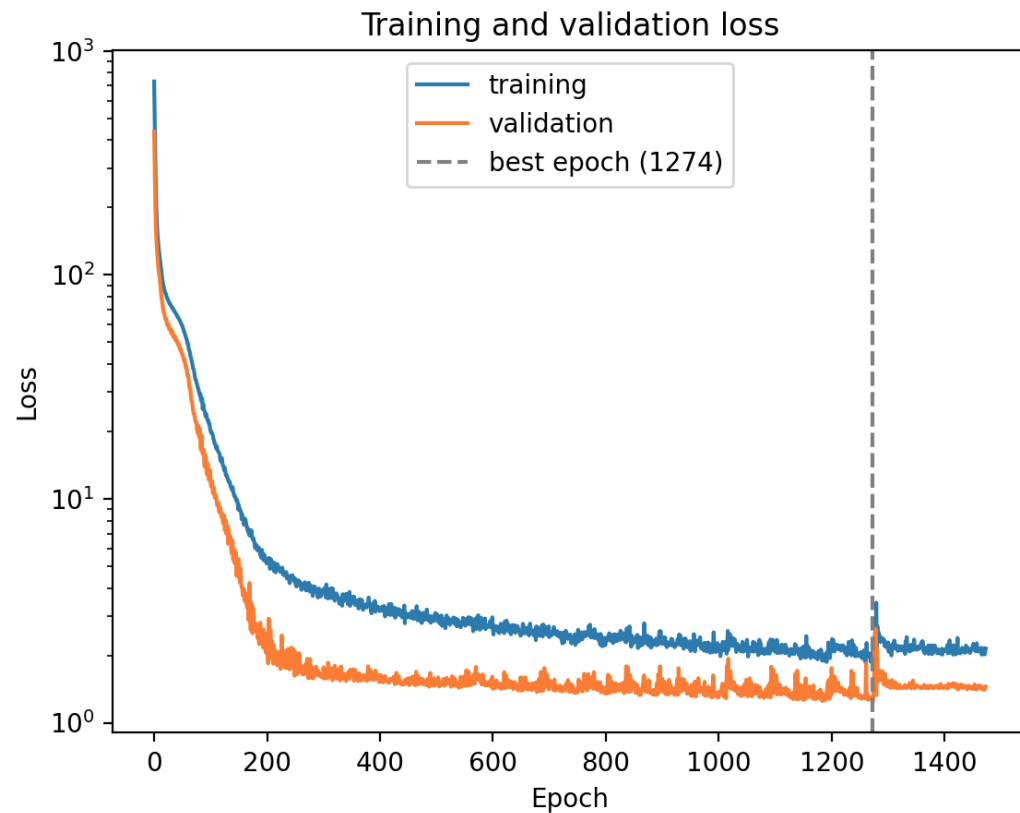
RESULTS: TEMPERATURE SCAN FOR FIRST SAMPLE



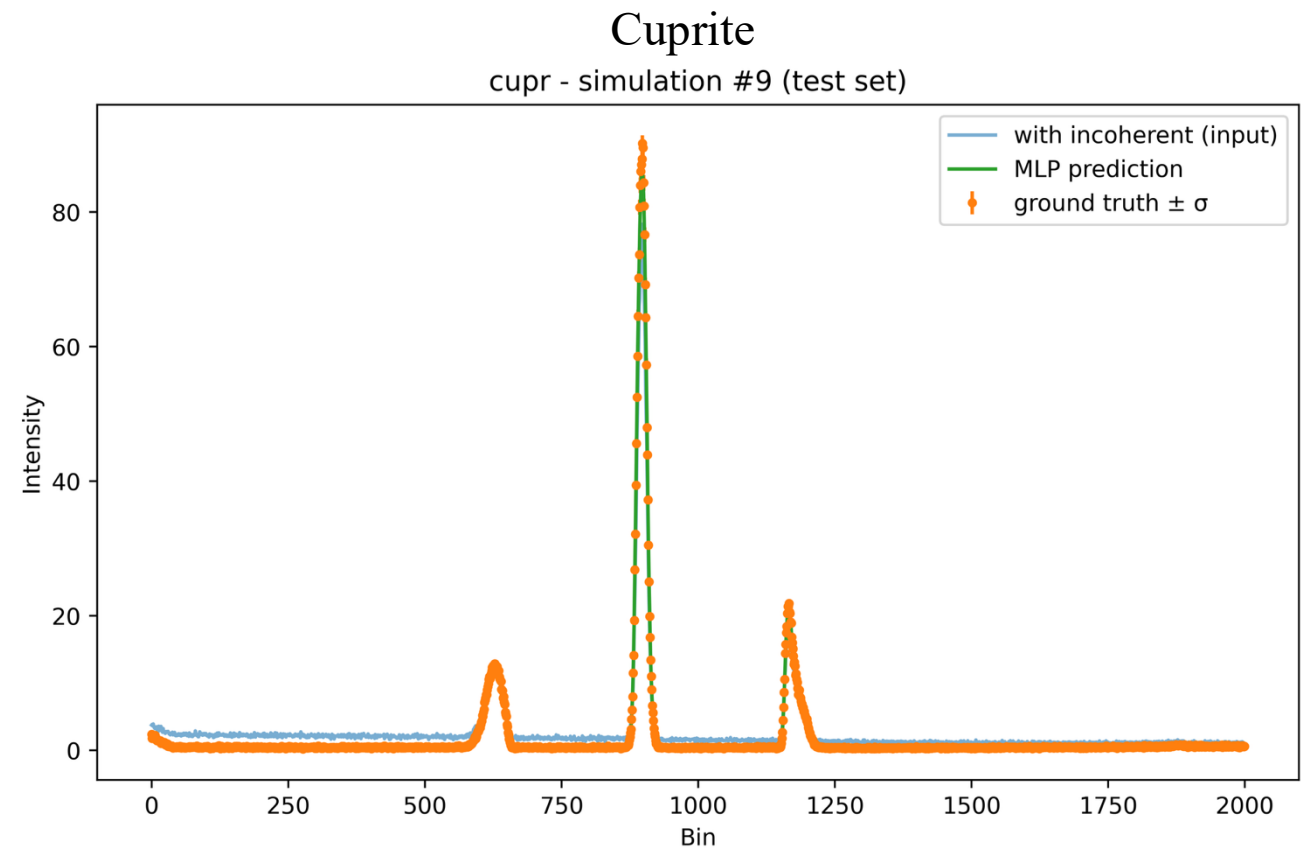
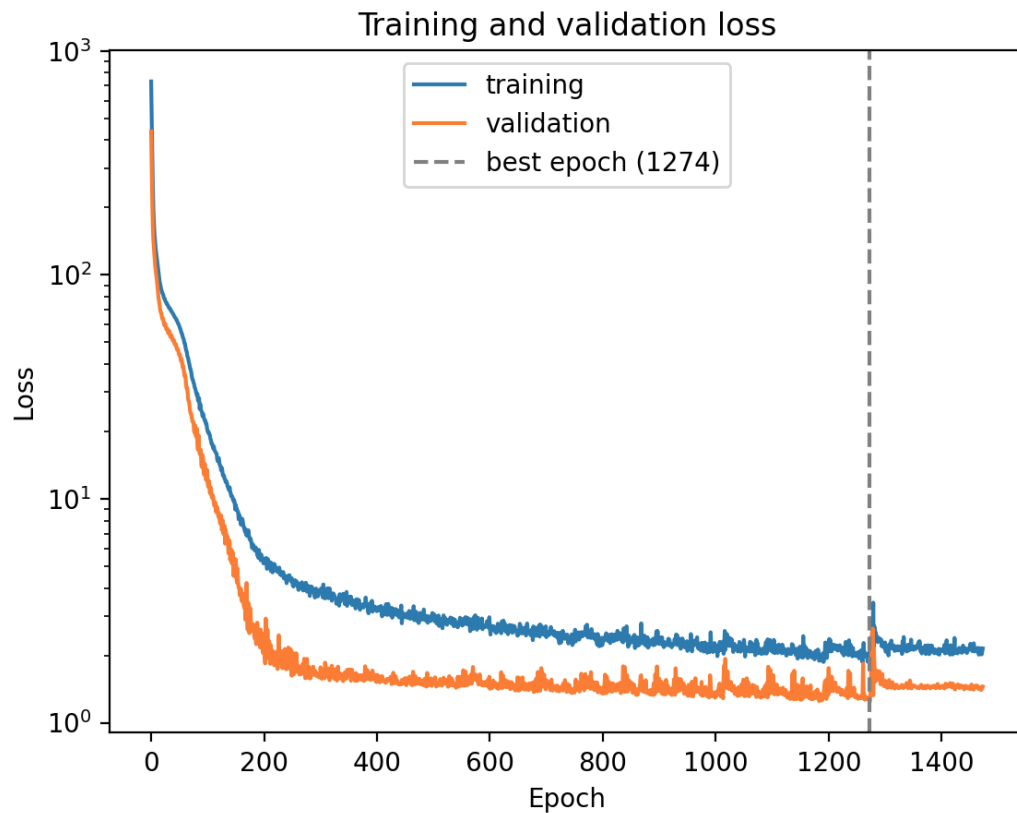
RESULTS: ANGLE ϑ SCAN FOR ALL 5 SAMPLES



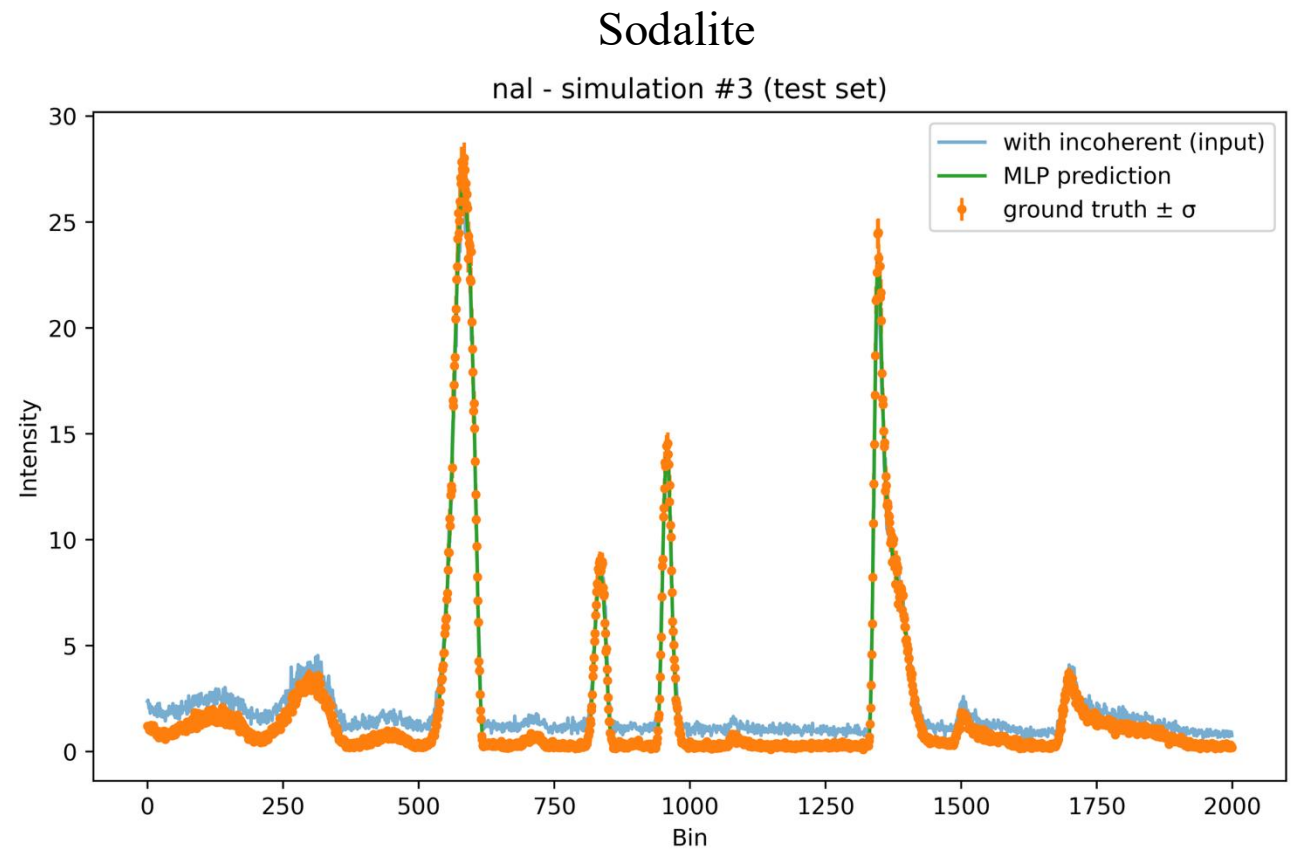
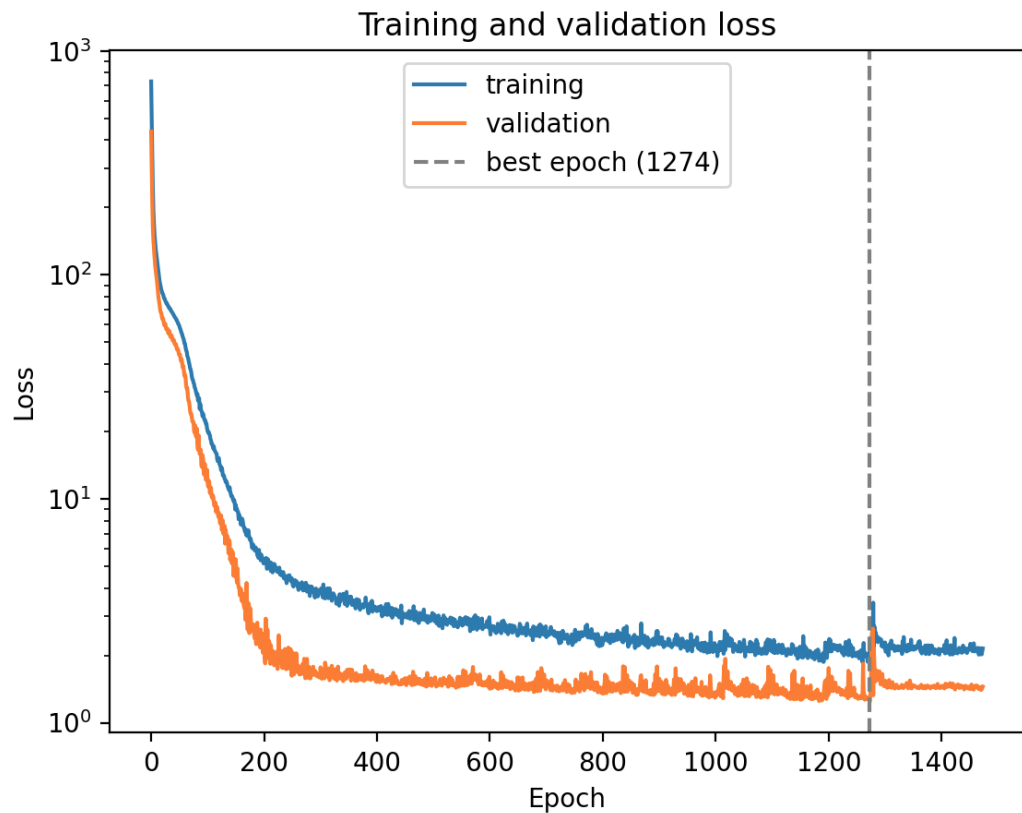
RESULTS: ANGLE ϑ SCAN FOR ALL 5 SAMPLES



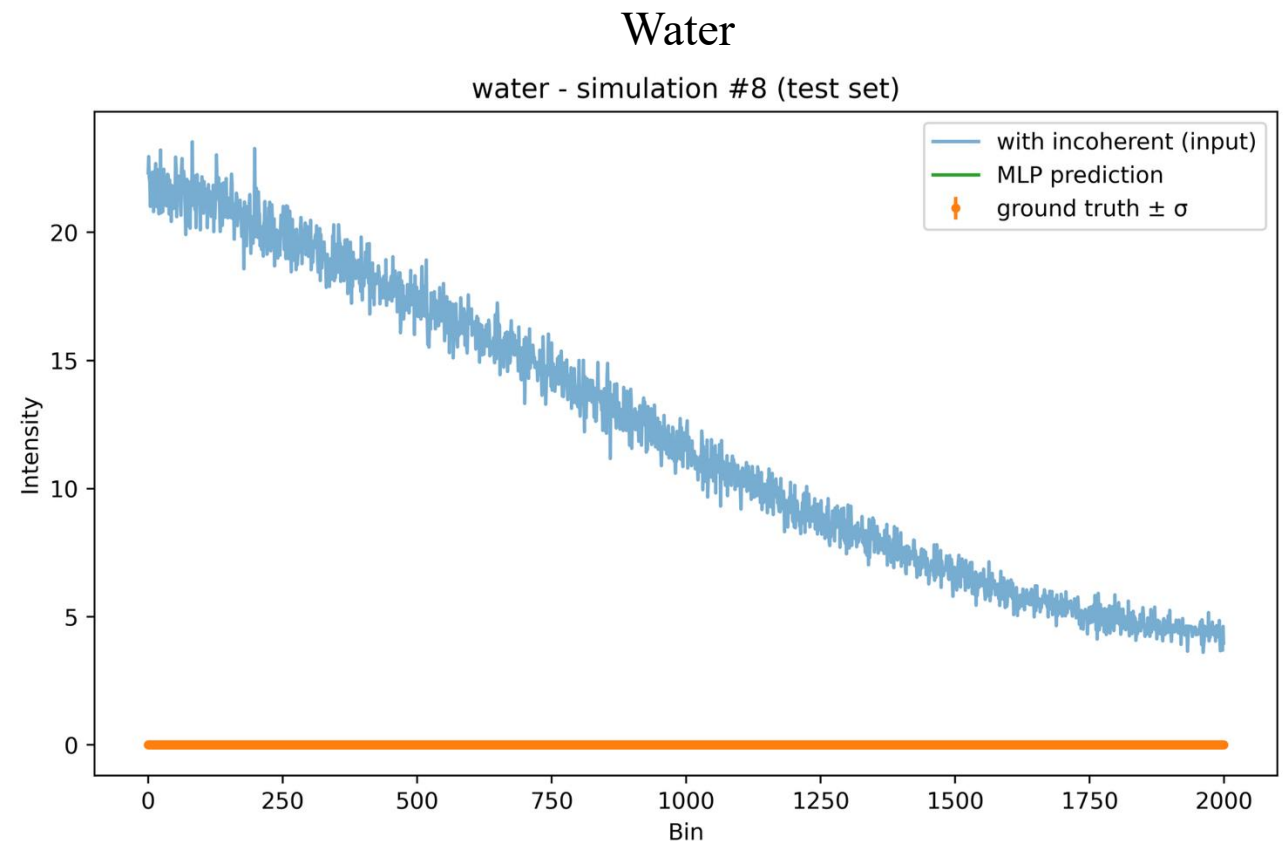
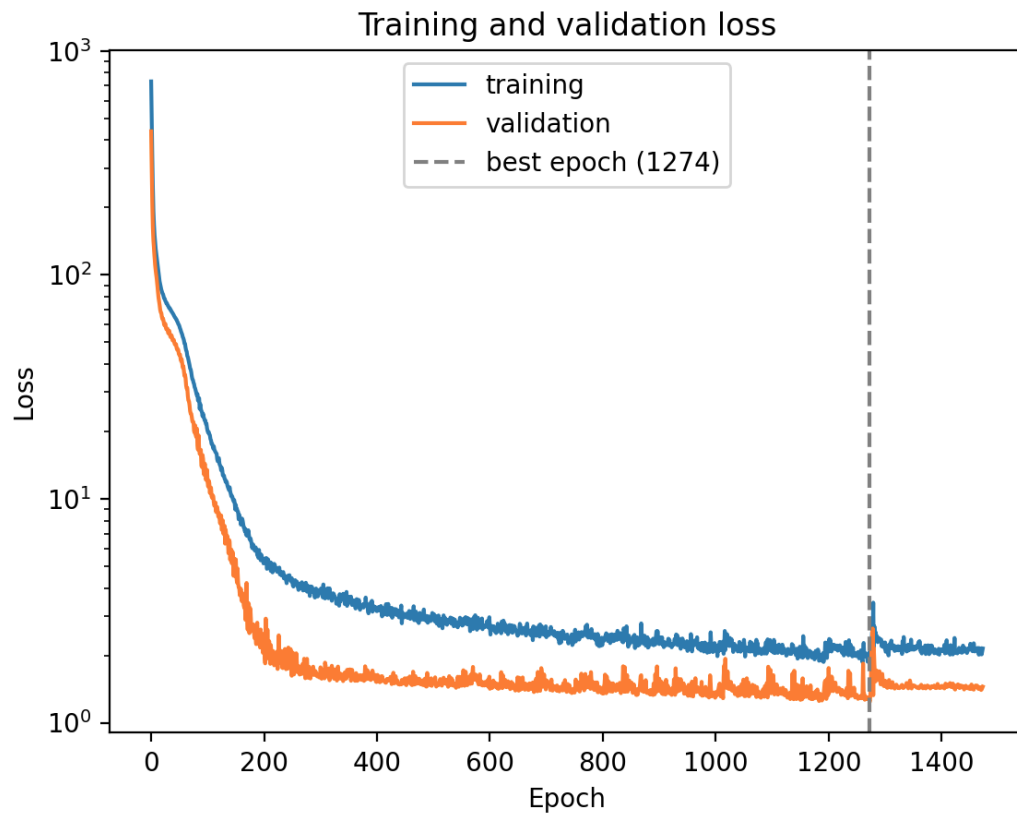
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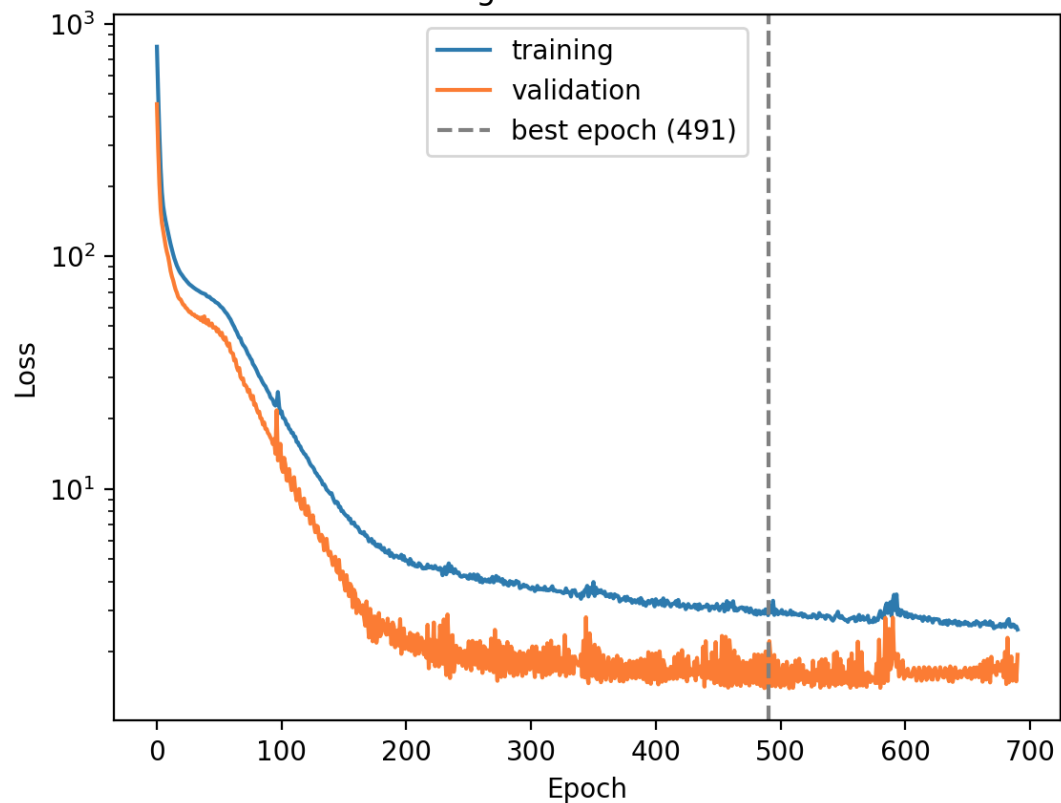


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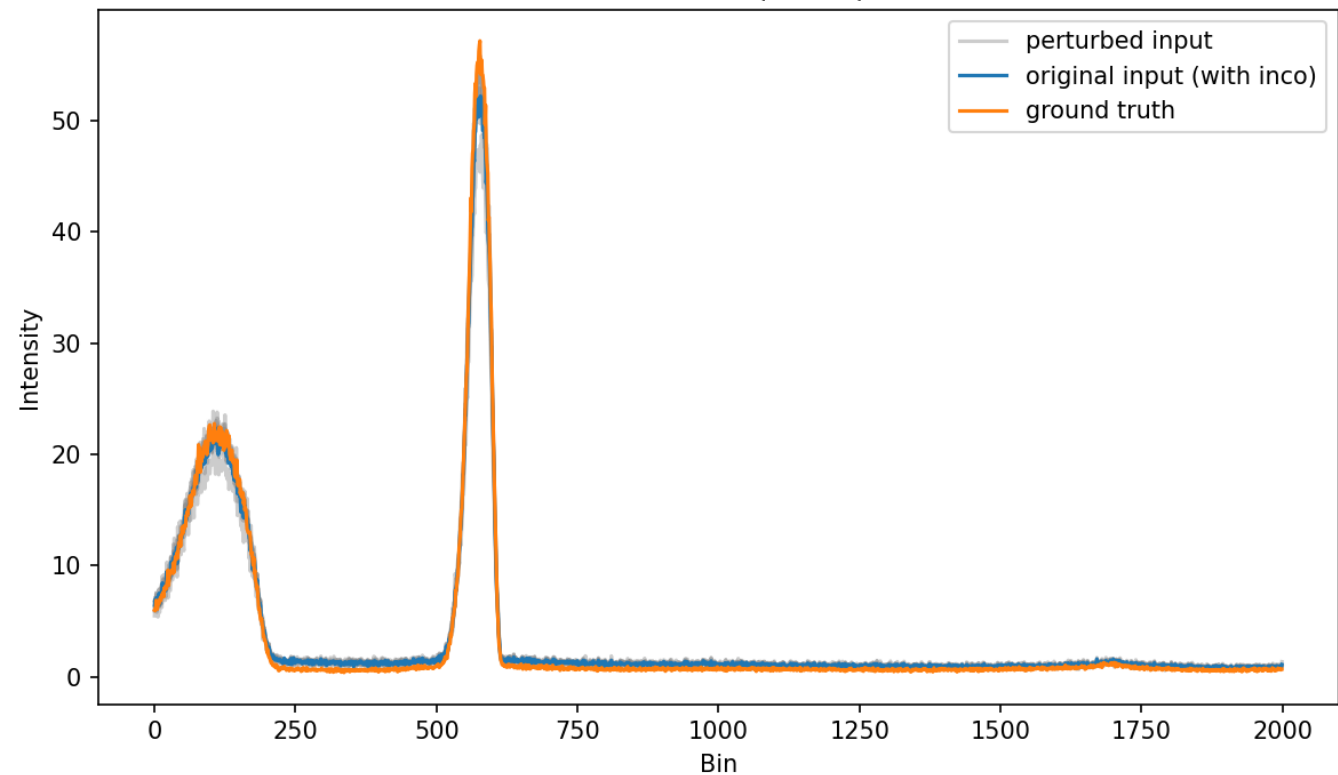


SHAKE IT UP: ADDING PERTURBATIONS

Training and validation loss



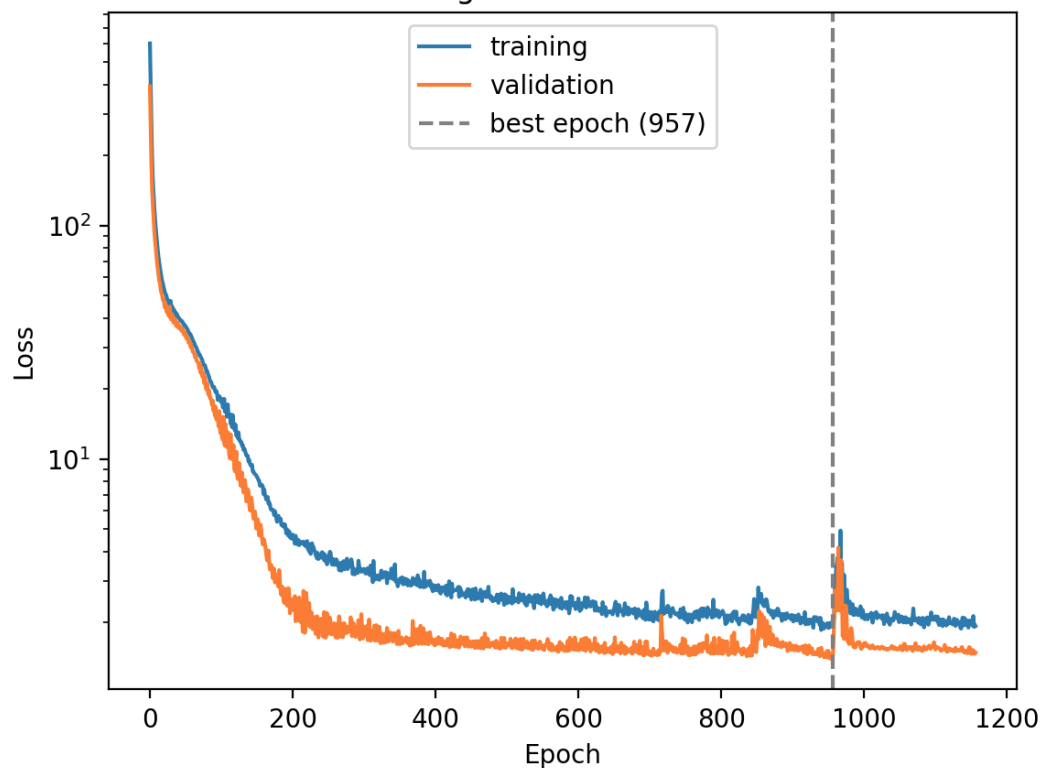
Perturbation example: cop #76



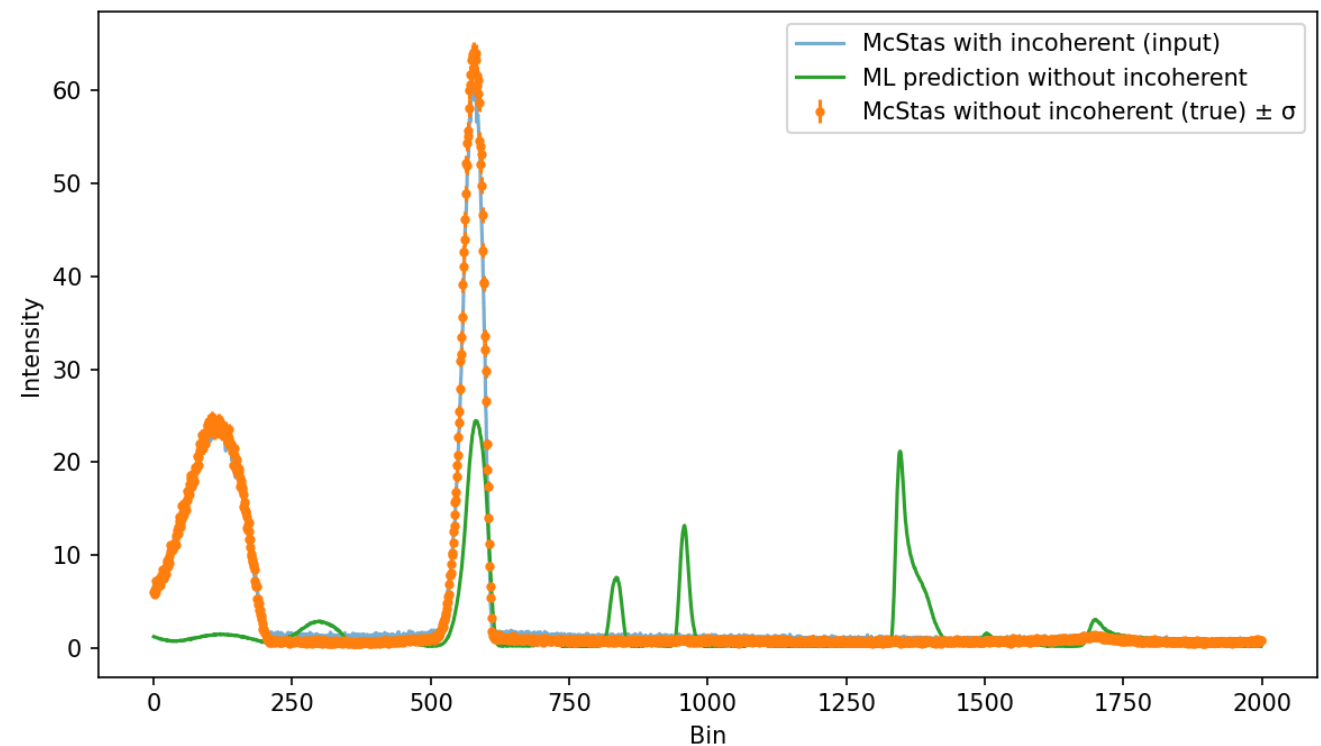
BLIND TEST: CAN THE MODEL HANDLE COPPER?

Mcstas simulations at different ϑ angles

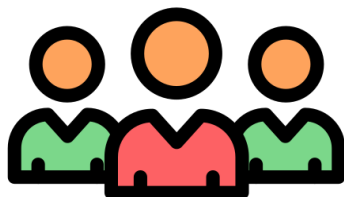
Training and validation loss



Prediction (unseen data): cop_inco 0



BLIND TEST: CAN THE MODEL HANDLE COPPER?



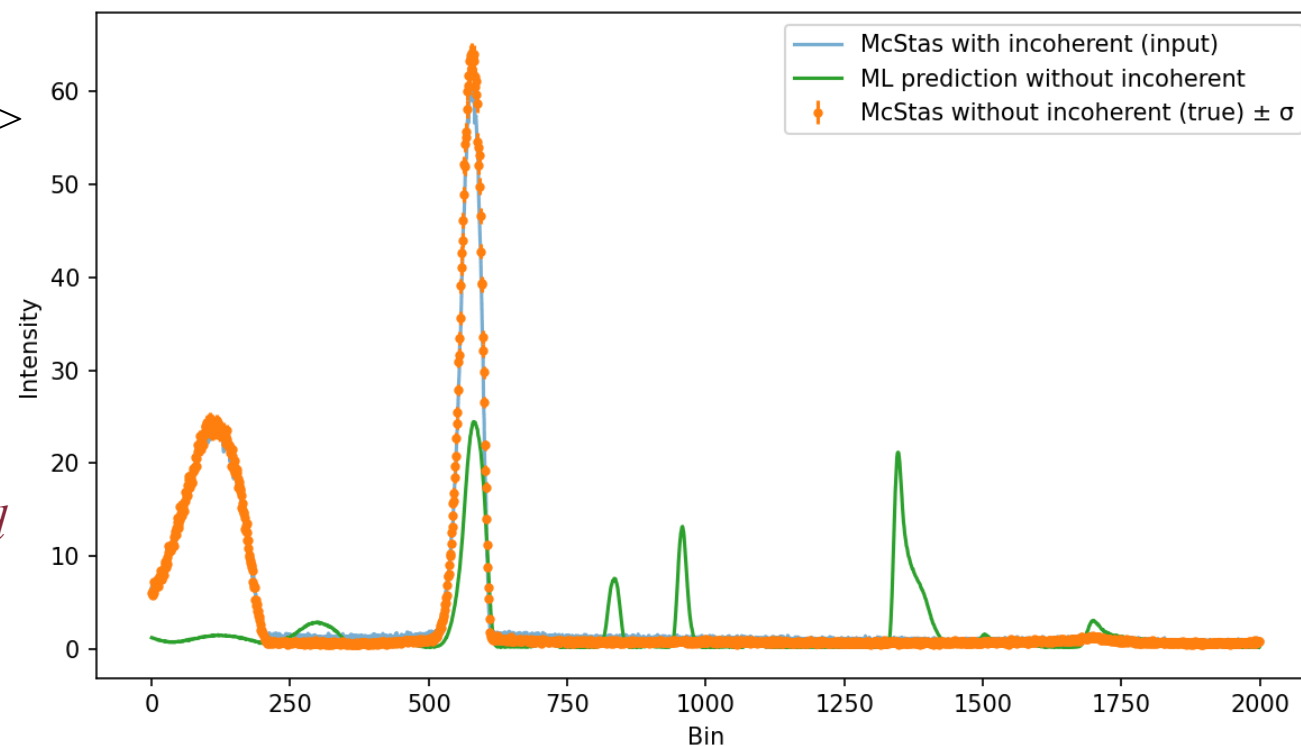
Us: << *Claude why the model fails?* >>

Claude: << *The model hasn't learned to remove the incoherent background. It has learned to recognize which of the four training materials a spectrum looks like and redraw that material's pattern. The extra peaks are most likely features copied from the other samples.* >>



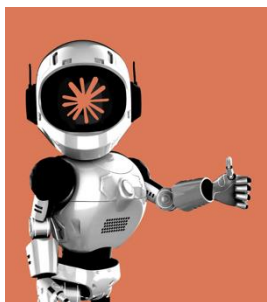
Mcstas simulations at different β angles for Copper

Prediction (unseen data): cop_inco 0



CLAUDE'S SUGGESTIONS

Claude: <<Changes that should help, most important first:



1. *Predict only the background. Change the model so its output is the input minus a predicted incoherent background, with the background kept positive. A model that does nothing then already gives the input back, which for cop is almost perfect. It can no longer lose real peaks or invent new ones.*
2. *Use a 1D convolutional network instead of the fully connected one. It looks at local shapes rather than memorising whole spectra, so it copes better with peaks in places it hasn't seen before.*
3. *Normalise each spectrum on its own, instead of dividing all of them by one shared maximum.*
4. *Add more materials to training if you can simulate them. Four is very few for a model that has to generalise to a new one. >>*

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SOLUTION 1: PREDICT ONLY THE BACKGROUND

Mcstas simulations at different β angles for Copper

Prediction (unseen data): cop_inco 0

